

Trevor Teolis

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Research Interests

Operator learning, machine learning for physics, bayesian inference, sampling, kinetic theory, interacting particle systems, collective dynamics.

Education

University of Illinois Chicago, Chicago, IL Ph.D., Mathematics, April 2025
Dissertation: *Well-posedness, long-time behavior, and multi-scale description of models of collective dynamics*. Advisor: Roman Shvydkoy.
University of Maryland, College Park, MD B.S., Mathematics and Computer Science, May 2019

Academic Positions

Rice University, Houston, TX Postdoctoral Researcher, Aug 2025–present
Advisor: Maarten V. de Hoop.
University of Illinois Chicago, Chicago, IL Doctoral Researcher, Aug 2020–May 2025
Advisor: Roman Shvydkoy.

Publications

1. R. Shvydkoy and T. Teolis. “Microscopic, mesoscopic, and macroscopic descriptions of the Euler Alignment System with adaptive communication strength.” *Discrete and Continuous Dynamical Systems*, 2025.
2. R. Shvydkoy and T. Teolis. “Well-posedness and long-time behavior of the Euler Alignment System with adaptive communication strength.” *Abel Symposium Proceedings*, 2024.
3. P. Nandori and T. Teolis. “Local Equilibrium of Particle Density in Planar Lorentz Processes.” *Nonlinearity*, 2021.

Preprints

1. S. D. Mis, T. Teolis, I. Dokmanić, M. Lassas, and M. V. de Hoop. “All Roads Lead to Markov Kernel Networks: A Representation Theorem for Discretization-Stable Learning.” Preprint, 2026.
2. T. Teolis and M. V. de Hoop. “In-context Maps as a Collision Operator: Stochastic Jump Transformer.” Preprint, 2026.
3. T. Teolis and M. V. de Hoop. “Target-adapted Green-Bessel SVGD: uniform-in-time propagation of chaos and last-iterate consistency.” arXiv:2609.08122, 2026.
4. S. Li, T. J. Maranzatto, J. Peszek, T. Teolis, S. Akkoc, K. Riedl, S. Ulukus, and N. García Trillos. “On the Diverse Dynamical Behaviors Arising in Deep Linear Transformers.” arXiv:2607.18584, 2026.
5. T. Teolis and M. V. de Hoop. “Riesz-Kernel Stein Variational Gradient Descent: Renormalized Entropy and Long-Time Particle Limits.” arXiv:2607.14527, 2026.
6. A. Balaji, T. Teolis, S. D. Mis, J. A. Lara Benitez, C. Wang, and M. V. de Hoop. “Hybrid operator learning of wave scattering maps in high-contrast media.” arXiv:2602.11197, 2026.
7. R. Shvydkoy and T. Teolis. “Long-time behavior of the Fokker–Planck–Navier–Stokes Alignment System.” arXiv:2508.07415, 2025.
8. A. Chertock, R. Shvydkoy, and T. Teolis. “Modulation of the monokinetic limit for models of collective dynamics.” arXiv:2508.05478, 2025.

Manuscripts in Preparation

1. T. Teolis, S. D. Mis, and M. V. de Hoop. “Hybrid Markov kernel networks: applications to wave modeling.”
2. T. Teolis, A. Siahkoohi, and M. V. de Hoop. “Finite-Mode Preconditioned SVGD with a Particle-Level Lyapunov Functional.”

Talks and Presentations

Invited Talks

- “Approximating Boltzmann solution maps by averaged measure-theoretic transformers,” minisymposium *Mathematical Foundations of Learning for Infinite-Dimensional Systems and PDEs*, SIAM Conference on Mathematics of Data Science, Salt Lake City, Utah, November 16–20, 2026. (upcoming)
- “Jump processes, stochastic in-context maps, mean-field limits, and the Boltzmann equation,” *Models of Emergence and Collective Dynamics*, 15th AIMS Conference, Athens, Greece, July 6–10, 2026.
- “Nanbu particles as stochastic in-context maps for Boltzmann dynamics,” *Measure Flows for Inverse Problems and Machine Learning*, SIAM Conference on Uncertainty Quantification, Minneapolis, Minnesota, March 22–25, 2026.

Contributed Talks

- Inverse Days 2026, Finnish Inverse Problems Society, University of Jyväskylä, Finland, December 14–17, 2026. (upcoming)
- “The s-model and the unconditional alignment of particles in a fluid,” Emergent Macroscopic Phenomena in Non-equilibrium Statistical Mechanics, GSSI, L’Aquila, Italy, May 2025.
- “A new model of collective dynamics with adaptive communication strength,” KinMAT Workshop, University of Warsaw, June 2024.
- “Proof techniques for alignment and flocking,” IDEAL Seminar (online), March 2024.

Research Presentations to Funders

- “FLOWERS: A warp drive for neural PDE solvers,” Oxy Annual Project Review Meeting, Houston, Texas, May 19, 2026.
- “Interacting particle systems in machine learning and inference,” TGS/GMIG Annual Project Review Workshop, Rice University, Houston, Texas, September 16, 2025.

Poster Sessions

- “The s-model and the unconditional alignment of particles in a fluid,” Workshop on Kinetic Theory and Fluids, UW–Madison, March 2025.
- “The s-model and the unconditional alignment of particles in a fluid,” Emergent Behavior in Complex Systems, IMSI, Chicago, March 2025.

Departmental Seminars

- “Long-time behavior and multi-scale description of models of collective dynamics,” Ph.D. Defense, UIC, April 2025.
- “A new model of collective dynamics with adaptive strength,” Analysis and Applied Math Seminar, UIC, February 2024.

Awards

- Victor Twersky Memorial Award, University of Illinois Chicago, 2024.
- Award for Graduate Research, University of Illinois Chicago, 2024.
- Travel Award, University of Illinois Chicago, 2024.

Workshops and Summer Schools

- Emergent Macroscopic Phenomena, GSSI, Italy, May 2025.
- Workshop on Kinetic Theory and Fluids, UW–Madison, March 2025.
- Emergent Behavior in Complex Systems, IMSI, Chicago, March 2025.
- Winter School: Boundary and Singularity in Fluid Mechanics, Stony Brook, January 2025.
- KinMAT Workshop, University of Warsaw, June 2024.
- Summer School: Fluid Dynamics, UMD, July 2023; Summer School: PDEs, Duke, August 2022.

Teaching

Rice University

- CMOR 422/522: Numerical Analysis, Co-Instructor, Fall 2026.
- CMOR 677: Mathematical Theory of Deep Learning, Co-Instructor, Spring 2026.

University of Illinois Chicago. Teaching Assistant for Calculus I–III, Differential Equations, and Linear Algebra, 2020–2025.

Professional Experience

Rampart Communications, Linthicum, MD

Developed and implemented polar-coded modulation schemes.

Summer Doctoral Intern, Jun–Aug 2024

Argonne National Laboratory, Lemont, IL

Developed algorithms for mixed-integer PDE-constrained optimization.

NSF Summer Intern, Jun–Aug 2021

National Institutes of Health, Bethesda, MD

Undergraduate Researcher and Software Engineer, Jan 2018–Jul 2020.

Built finite-element models of brain tumor growth using MRI and DTI data.